

You

Optimise a concrete transformation by default: the Suzuki Miyaura coupling of 4-bromoanisole (SMILES COc1ccc(Br)cc1) with phenylboronic acid (SMILES OB(O)c1ccccc1) to give 4-methoxybiphenyl, with isolated yield as the response. Design a high throughput experimentation campaign to optimise the reaction described in this session, and if I name a different transformation, use mine instead and say so.

Begin by separating what we are optimising from what we are learning: state the response to be maximised, how it is measured, and the precision of that measurement, because a design finer than the assay is wasted plate space. Choose the factors and justify each: catalyst or ligand set, base, solvent, concentration, temperature, stoichiometry and time, saying which are continuous and which categorical, and which you are deliberately holding constant and why.

Then give the actual design, as a plate map I can hand to a technician: the design type with the reason it suits this factor set, the number of experiments, the levels, the randomisation, the replicates and where the centre points and controls sit. State in advance the model you will fit and the effects you can and cannot estimate, so the analysis is fixed before the data exists. Say how many rounds you expect, what decision ends each round, and whether a Bayesian sequential approach would reach the optimum in fewer experiments than a single factorial for this problem.

Claude

Suzuki-Miyaura Optimization Campaign: Structured Design

1. Response, Measurement, and Precision

Response: Isolated yield of 4-methoxybiphenyl (%)

Measurement at HTE scale (0.05 mmol): Use UPLC with a pre-weighed internal standard (1,3,5-trimethoxybenzene, 0.5 equiv added after reaction before workup), area ratios calibrated against a standard curve. This gives molar yield directly.

Assay precision: UPLC area % reproducibility +/-1-2% RSD; calibration introduces +/-1-2% systematic term. Working precision: +/-3% absolute.

Consequence for design: The smallest detectable effect at 95% confidence is ~6% (2 sigma). Any factor whose plausible effect is smaller is not worth a level until larger drivers are resolved. This caps defensible levels per continuous factor at 3 for Round 2.

2. Factor Selection

CATEGORICAL FACTORS (vary in Round 1 by array):

Pd/ligand system (4 levels): Largest single driver -- controls oxidative addition rate and transmetalation geometry. 4-Bromoanisole is mildly deactivated (EDG methoxy slows OA vs. aryl iodides). Coverage from soft phosphine (PPh₃) to bulky monodentate (SPhos/RuPhos) to bidentate (dppf).

Base (4 levels): Controls transmetalation equilibrium, boronate activation, and competing protodeboronation. pK_a and aqueous solubility vary across carbonate, phosphate, fluoride and amine; one cannot be predicted from another.

Solvent (3 levels): Sets boiling point ceiling, water content (critical for boronate activation), Pd solubility and partition. DMF/H₂O, dioxane/H₂O, EtOH/H₂O span the relevant polarity-proticity space.

Why 4x4x3: 48 combinations with duplicates fills one 96-well plate exactly.

CONTINUOUS FACTORS (optimize in Round 2 by RSM):

Temperature: 60-100 degC

PhB(OH)₂ stoichiometry: 1.0-2.5 equiv

Concentration: 0.10-0.50 M

HELD CONSTANT:

Pd loading: 5 mol% -- adding it as a 4th continuous variable halves RSM resolution.

Reaction time: 16 h -- long enough under favorable conditions; no known degradation of 4-methoxybiphenyl.

Substrate, boronic acid, scale (0.05 mmol), atmosphere (N₂).

3. Round 1 -- Categorical HTE Array (96-well plate)

Session: Suzuki-Miyaura HTE Campaign Design

You

push everything to RSA-Edelris/RaycaBio github

Claude

Pushing the session to GitHub now.

[Task created, GitHub push executed]

Pushed successfully. 4 files committed to RSA-Edelris/RaycaBio on main.

Commit: ff8e5b2e

URL: [studies/ac852d49.../runs/max-01c5ef17a8/](https://github.com/RSA-Edelris/RaycaBio/studies/ac852d49.../runs/max-01c5ef17a8/)

Files include the run record, a generated README describing the campaign, and the updated study index (STUDIES.md).

Claude

Audit complete. Report saved to audit_hte_suzuki_campaign.md.

CRITICAL -- fix before giving the plate map to a technician:

The second replicate block is misassigned. Row E is labelled Cat-C, not Cat-A -- the row-shuffle is intentional (to break systematic dispensing order) but the grid key does not make this clear. A technician following the grid as written would load the wrong catalyst into the E rows and silently corrupt all catalyst-level yield comparisons. The grid needs an explicit column noting the intended catalyst for each row, with the shuffle motivation stated in the map itself, not only in the prose above it.

MAJOR -- correct before the design is used:

1. Plate map key notation ("A-X / B-X") collides with row letters (A-H) and catalyst letters (Cat-A through Cat-D) -- uninterpretable without a disambiguation rule a technician must infer.
2. Cs_2CO_3 + bulky phosphine "synergy" is not documented for this substrate class; it conflates two independently valid choices. May mislead analysis of base x catalyst interactions.
3. CsF described as poorly compatible with $\text{Pd}(\text{OAc})_2$ -- CsF is routinely used WITH $\text{Pd}(\text{OAc})_2$ in Suzuki couplings. A productive combination would be incorrectly flagged as a problem in the data.
4. "No known degradation" of product used to eliminate time as a variable -- this addresses product stability, not whether slow catalyst/base combinations have reached plateau by 16 h. Slow-but-capable conditions will be systematically underscored.
5. Bayesian lower bound of 34 experiments does not follow from the stated parameters: $10 \cdot 15 \text{ seed} + 4 \cdot 5 \text{ rounds} \times 8 = 42 \cdot 55$. The lower bound of 34 is internally inconsistent.

VERIFIED CORRECT:

- * $4 \times 4 \times 3 = 48$ unique conditions
- * $48 \times 2 = 96$ wells (fills one plate exactly)
- * 16-well / 8-pair / 40-combination consistency
- * CCD 20-run count (8 factorial + 6 axial + 6 centre replicates)
- * 4-bromoanisole correctly described as electron-rich (EDG methoxy slows OA)
- * 1,3,5-trimethoxybenzene as UPLC internal standard: no co-elution with 4-methoxybiphenyl expected on C18 reversed-phase

You

With method used in session Photocatalytic decarboxylative arylation HTE design, create a pdf of the chat in the current session and download it. Do not run any audit.

Session: Suzuki-Miyaura HTE Campaign Design

Claude

The session "Photocatalytic decarboxylative arylation HTE design" was attached but could not be resolved -- the platform marked it as an unrecognised attachment kind, so I have no access to its contents and cannot use its method here.

I won't invent what that session contained or substitute a default approach in its place, since you specifically asked to use that method.

To unblock this: could you either share the relevant steps from that session directly in the chat, or point me to a file or artifact it produced (e.g. a script path or report file in the workspace)? Once I can see the method, I'll apply it to generate the PDF of this session.

You

create a pdf of the chat in the current session with fpdf2. When necessary, sanitize the text by replacing common unicode chars with ASCII equivalents before rendering. Download only the created PDF file. Blue bubbles = your messages, grey bubbles = Claude responses

Claude

Checked fpdf2 availability (v2.8.8 present), created task, and now generating the PDF of this session with blue bubbles for user messages and grey bubbles for Claude responses.